Notes and Solutions

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0.1 Maxwell's Equations

Definition 0.1.1 (Planck length)**.**

$$l_P = \sqrt{\frac{\hbar\kappa}{c^3}}, \tag{1}$$

where $\hbar$ is Planck's constant, κ is Newton's gravitational constant, and c is the speed of light.

Definition 0.1.2 (Maxwell's equations)**.** Let $\vec{E}$ be the electric field, $\vec{B}$ be the magnetic field, ρ be the charge density, and $\vec{j}$ be the current density.

$$\begin{aligned}
&\nabla \cdot \vec{B} = 0 && \nabla \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = 0 && \\
&\nabla \cdot \vec{E} = \rho && \nabla \times \vec{B} - \frac{\partial \vec{E}}{\partial t} = \vec{j}, && \text{or equivalently} \\
&\nabla \cdot \vec{\mathcal{E}} = \rho && \nabla \times \vec{\mathcal{E}} = i\left(\frac{\partial \vec{\mathcal{E}}}{\partial t} + \vec{j}\right), && \text{where } \vec{\mathcal{E}} = \vec{E} + i\vec{B},
\end{aligned}$$

where $\nabla\cdot = \text{div}$ is the divergence and $\nabla\times$ is the curl. In the vacuum, $\rho = 0$ and $\vec{j} = 0$.

0.2 Manifolds

Definition 0.2.1 (topological space)**.** A topological space is a set X together with a family of subsets of X, called open sets, required to satisfy the following conditions:

1. The empty set and X itself are open,
2. If U, V are open, so is $U \cap V$,
3. If the sets $U_\alpha \subset X$ are open, so is the union $\cup U_\alpha$.

The collection of sets taken to be open is called the topology of X. An open set containing a point $x \in X$ is called a neighborhood of x. The complement of an open set is called closed.

Definition 0.2.2 (continuity)**.** A function $f : X \to Y$ from one topological space is defined to be continuous if, given any open set $U \subset Y$, the inverse image $f^{-1}U \subset X$ is open.

Definition 0.2.3 (epsilon-delta continuity)**.** A function $f : \mathbb{R}^n \to \mathbb{R}^m$ is continuous if, for all $x \in \mathbb{R}^n$ and all $\varepsilon > 0$, there exists $\delta > 0$ such that $||y - x|| < \delta$ implies $||f(y) - f(x)|| < \varepsilon$.

Definition 0.2.4 (chart)**.** Given a topological space X and an open set $U \subset X$, we define a chart to be a continuous function $\varphi : U \to \mathbb{R}^n$ with a continuous inverse $\varphi^{-1} : \varphi(U) \to U$, and we define a transition function to be a composition of the form $\varphi_\alpha \circ {\varphi_\beta}^{-1}$.

Definition 0.2.5 (atlas)**.** An atlas is a collection of charts $\varphi_\alpha : U_\alpha^n$, where U_α are open sets, such that the transition functions are continuous where they are defined.

Definition 0.2.6 (topological n-manifold)**.** An n-dimensional manifold is a topological space M equipped with an atlas whose charts cover it.

Definition 0.2.7. A smooth manifold is a topological manifold whose transitions functions are smooth where they are defined.

Definition 0.2.8 (induced topology)**.** Given a topological space X and a subset $S \subset X$, define the induced topology on S to be the topology in which the open sets are of the form $U \cap S$, where U is open in X.

Definition 0.2.9 (n-sphere)**.** Let S^n be the unit sphere in $\mathbb{R}^{n+1}$:

$$S^n = \left\{ \vec{x} \in \mathbb{R}^{n+1} \mid \sum_{i=1}^{n+1} (x^i)^2 = 1 \right\}. \tag{2}$$

Definition 0.2.10 (product topology)**.** Given topological spaces X and Y, we give $X \times Y$ the product topology in which a set is open if and only if it is a union of sets of the form $U \times V$, where U is open in X and V is open in Y.

Definition 0.2.11 (disjoint union topology)**.** Given topological spaces X and Y, we give $X \cup Y$ the disjoint union topology in which a set is open if and only if it is the union of an open subset of X and an open subset of Y.

Definition 0.2.12 (compact space)**.** A topological space X is compact if for every cover of X by open sets U_α there is a finite collection $U_{\alpha_1}, \ldots, U_{\alpha_n}$ that covers X.

Definition 0.2.13 (compact manifold)**.** A manifold M is compact if and only if every sequence in M has a convergent subsequence.

0.3 Vector Fields

Definition 0.3.1 ($C^\infty(M)$ algebra over $\mathbb{R}$)**.** The set $C^\infty(M)$ of smooth (real-valued) functions on a manifold M 'having infinitely many continuous derivatives' is an algebra over the real numbers, meaning that it is closed under (pointwise) addition and multiplication, as well as multiplication by real numbers, and the following batch of rules holds:

additive commutativity:	$f + g = g + f$
additive associativity:	$f + (g + h) = (f + g) + h$
multiplicative associativity:	$f(gh) = (fg)h$
right-distributive over addition:	$f(g + h) = fg + fh$
left-distributive over addition:	$(f + g)h = fh + gh$
unital:	$1f = f$
scalar left-distributive:	$\alpha(\beta f) = (\alpha\beta)f$
scalar right-distributive over addition:	$\alpha(f + g) = \alpha f + \alpha g$
right-distributive over scalar addition:	$(\alpha + \beta)f = \alpha f + \alpha f.$

Furthermore, it is a commutative algebra because $fg = gf$.

Definition 0.3.2 (module ovector field)**.** A vector field $v \in \mathrm{Vect}(M)$ is a function $v : C^\infty(M) \to C^\infty(M)$ satisfying the following properties:

additivity:	$v(f + g) = v(f) + v(g)$	
multiplicativity:	$v(\alpha f) = \alpha v(f)$	
product/Leibniz law:	$v(fg) = v(f)g + fv(g),$	and
addition:	$(v + w)f = v(f) + w(f),$	
multiplication:	$(gv)(f) = gv(f).$	

In fact, $\mathrm{Vect}(M)$ is a module over $C^\infty(M)$.

Definition 0.3.3 (basis)**.** The vector fields $\{\partial_\mu\}$ on $\mathbb{R}^n$ form a basis of $\mathrm{Vect}(\mathbb{R}^n)$ because they are linearly independent and span $\mathrm{Vect}(\mathbb{R}^n)$ as a module over $C^\infty(M)$, i.e. every vector field on $\mathbb{R}^n$ is a linear combination of the following form:

$$v^\mu \partial_\mu = v^1 \partial_1 + \cdots + v^n \partial_n, \tag{3}$$

for components $v^\mu \in C^\infty$.

Definition 0.3.4 (tangent vector space)**.** We define a tangent vector at $p \in M$ to be a function $v_p \in C^\infty(M) \to \mathbb{R}$ satisfying the following three properties:

$$\begin{aligned}
&\text{addition:} && v_p(f+g) = v_p(f) + v_p(g) \\
&\text{multiplication:} && v_p(\alpha f) = \alpha v_p(f) \\
&\text{product/Leibniz law:} && v_p(fg) = v_p(f)g(p) + f(p)v_p(g),
\end{aligned}$$

Let T_pM denote the tangent space of all tangent vectors at p.

Definition 0.3.5 (curve)**.** A curve is a smooth function $\gamma : \mathbb{R} \to M$, i.e. one where $f(\gamma(t))$ depends smoothly on t for any $f \in C^\infty(M)$.

Definition 0.3.6 (pullback)**.** The pullback of $f : N \to \mathbb{R}$ from N to M by $\phi : M \to N$ is the following composite:

$$\phi^\star f = f \circ \phi, \tag{4}$$

and we say that f is contravariant.

Definition 0.3.7 (smooth map)**.** We say $\phi : M \to N$ is a (smooth) map if $f \in C^\infty(N)$ implies that $\phi^\star f \in C^\infty(M)$.

Definition 0.3.8 (pushforward)**.** The pushforward $\phi_\star v \in T_{\phi(p)}M$ of $v \in T_pM$ by $\phi : M \to N$ is defined below:

$$(\phi_\star v)(f) = v(\phi^\star f), \tag{5}$$

and we say that v is covariant.

Definition 0.3.9 (integral curve)**.** The integral curve $\gamma : \mathbb{R} \to M$ through $p \in M$ of the vector field $v \in \mathrm{Vect}(M)$ satisfies $\gamma(0) = p$ and $\gamma'(t) = v_{\gamma(t)}$.

Definition 0.3.10 (integrable)**.** We say that the vector field v is integrable if all the integral curves are defined for all t.

Definition 0.3.11 (flow generation)**.** Let $\phi_t : M \to M$ be the smooth integral curve of v through $p \in M$. The flow generated by v, i.e. the family of maps $\{\phi_t\}$, is defined below:

$$\frac{d}{dt}\phi_t(p) = v_{\phi_t(p)}. \tag{6}$$

Definition 0.3.12 (Lie bracket)**.** Given $v, w \in \mathrm{Vect}(M)$, the Lie bracket $[v, w] = vw - wv$ is defined by the following commutator:

$$[v, w](f) = v(w(f)) - w(v(f)), \qquad \forall f \in C^\infty(M), \tag{7}$$

and we say the vector fields commute if $[v, w] = 0$.

0.4 Differential Forms

Remark 1 (linearity of directional derivative)**.** *The directional derivative operator $v \mapsto \nabla f \cdot v$ is linear over $C^\infty(\mathbb{R}^n)$ since $\nabla f \cdot (gv) = g(\nabla f \cdot v)$.*

Definition 0.4.1 (1-form)**.** We define a 1-form $\omega \in \Omega^1(w)$ on any manifold M to be a map from $\mathrm{Vect}(M)$ to $C^\infty(M)$ that is linear over $C^\infty(M)$:

$$\omega(v+w) = \omega(v) + \omega(w)\omega(gw) \qquad\qquad = g\omega(w)$$

Definition 0.4.2 (exterior derivative)**.** We call the 1-form $df(v) = \delta f \cdot v = vf$ the differential of f, or the exterior derivative of f.

The map $d : C^\infty(M) \to \Omega^1(M)$ that sends each function f to its differential df is also called the differential, or exterior derivative.

Definition 0.4.3 (Kronecker delta)**.** The Kronecker delta $\delta^\mu_\nu = 1$ if $\mu = \nu$ and 0 otherwise.

Definition 0.4.4 (cotangent vector)**.** Given $p \in M$, a cotangent vector at p is defined to be a linear map $\omega : T_pM \to \mathbb{R}$. Let $T^\star_pM$ denote the space of all cotangent vectors at p. Given a 1-form $\omega \in \Omega^1(M)$, we can define a cotangent vector $\omega_p \in T^\star_pM$ as follows:

$$w_p(v_p) = \omega(v)(p). \tag{8}$$

Definition 0.4.5 (dual vector space)**.** Given any vector space V, the dual vector space $V^\star$ is defined to be the space of all linear functionals $\omega : V \to \mathbb{R}$.

In particular, the cotangent space $T^\star_pM$ is the dual of the tangent space.

More generally, linear maps $f : V \to W$ yield a dual map $f^\star : W^\star \to V^\star$ defined by $(f^\star\omega)(v) = \omega(f(v))$.

Definition 0.4.6 (cotangent pullback)**.** If ω is a cotangent vector at $\phi(x)$, we define the pullback of ω by ϕ as follows:

$$(\phi^\star\omega)(v) = \omega(\phi_\star v),$$

or as a global 1-form on M given a 1-form ω on $N = \mathrm{codomain}(\phi)$ as follows:

$$(\phi^\star\omega)_p = \phi^\star(w_q),$$

where $q = \phi(p)$.

Definition 0.4.7 (exterior derivative naturality)**.** Given a function $f \in C^\infty(N)$ and a map $\phi : M \to N$, we have the following naturality:

$$\phi^\star(df) = d(\phi^\star f).$$

Definition 0.4.8 (local coordinates)**.** We can use φ to pull back the coordinate functions $x^\mu : \mathbb{R}^n \to \mathbb{R}^n$ to local coordinates $x^\mu = \varphi^\star x^\mu : U \to \mathbb{R}^n$ on U. Any function on U can be written as a function $f(x^1, \ldots, x^n)$ of these local coordinates.

Definition 0.4.9 (coordinate vector fields)**.** Similarly, we can use φ^{-1} to push forward the basis of coordinate vector fields $\partial_\mu \in \mathrm{Vect}(\mathbb{R}^n)$ to the coordinate vector fields $\partial_\mu = \varphi^{-1}_\star\partial_\mu \in \mathrm{Vect}(M)$ associated to the local coordinates x^μ on U. One thus writes any vector field v on U as $v = v^\mu\partial_\mu$.

Definition 0.4.10 (coordinate 1-forms)**.** Similarly, we can use φ to pull back the basis of coordinate 1-forms $dx^\mu \in \Omega^1(\mathbb{R}^n)$ to the coordinate 1-forms $dx^\mu = \phi^\star dx^\mu \in \Omega^1(U)$ associated to the local coordinates x^μ on U. We can write any 1-form $\omega \in \Omega^1(U)$ as $\omega = \omega_\mu dx^\mu$.

Definition 0.4.11 (active/passive transformations)**.** A passive coordinate transformation is a change of coordinate functions. An active coordinate transformation is a diffeomorphism $\phi : M \to M$.

Definition 0.4.12 (components)**.** The components of $v = v^\mu e_\mu$ are the functions $v^1, \ldots, v^n$on U.

Definition 0.4.13 (exterior algebra of forms)**.** The exterior algebra over V, denoted ΛV, is the algebra generated by V with the relations

$$v \wedge w = -w \wedge v$$

for all $v, w \in V$. In general, for any vector space V, we define $\Lambda^p V$ to be the subspace of ΛV consisting of linear combinations of p-fold products of vectors in V. Elements of $\Lambda^p V$ are said to have degree p.

Definition 0.4.14 (direct sum)**.** Recall that a vector space V is a direct sum of subspaces $L_1, \ldots, L_n$ if every vector $v \in V$ can be uniquely expressed as $v_1 + \cdots + v_n$, where $v_i \in L_i$. Given vector subspaces $V_1, \ldots, V_n$, their direct sum $V_1 \oplus \cdots \oplus V_n$, sometimes written

$$\bigoplus_{i=1}^{n} V_i,$$

is defined as the vector space of all n-tuples $(v_1, \ldots, v_n)$ with $v_i \in V_i$, where addition and scalar multiplication are defined componentwise.

Definition 0.4.15 (differential forms)**.** We define the differential forms on M, denoted $\Omega(M)$, to be the algebra over $C^\infty(M)$ generated by $\Omega^1(M)$ with the relations

$$\omega \wedge \mu = -\mu \wedge \omega,$$

for all $\omega, \mu \in \Omega^1(M)$. We allow all locally finite linear combinations, tht is, those for which every point $p \in M$ has a neighborhood where only finitely many terms are nonzero.

Exercises

0.5 GFKG: Maxwell's Equations

1. *Proof (electromagnetic plane wave).* Let $\vec{k}$ be a vector in $\mathbb{R}^3$ and let $\omega = |\vec{k}|$. Fix $\vec{E} \in \mathbb{C}^3$ with $\vec{k} \cdot \vec{E} = 0$ and $i\vec{k} \times \vec{E} = \omega \vec{E}$. We will show that

$$\vec{\mathcal{E}}(t, \vec{x}) = \vec{E} e^{-i(\omega t - \vec{k} \cdot \vec{x})}, \qquad \text{for } \vec{k} = \begin{pmatrix} k^x \\ k^y \\ k^z \end{pmatrix} \text{ and } \vec{E} = \begin{pmatrix} E^x \\ E^y \\ E^z \end{pmatrix},$$

satisfies the vacuum Maxwell equations: $\nabla \cdot \vec{\mathcal{E}} = 0$ and $\nabla \times \vec{\mathcal{E}} = i \dfrac{\partial \vec{\mathcal{E}}}{\partial t}$.

We compute the partial derivative with respect to time below:

$$\frac{\partial \vec{\mathcal{E}}}{\partial t} = \vec{E} e^{-i(-\vec{k} \cdot \vec{x})} \frac{\partial}{\partial t} \left(e^{-i(\omega t)} \right) = \vec{E} e^{-i(-\vec{k} \cdot \vec{x})} \left(-i\omega e^{(-i\omega)t} \right) = -i\omega \vec{\mathcal{E}}.$$

and of a spatial component $\mu = x, y,$ or z with respect to a spatial component z below:

$$\begin{aligned} \frac{\partial \mathcal{E}^\mu}{\partial z} &= E^\mu e^{-i(\omega t - k^x x - k^y y)} \frac{\partial}{\partial z} \left(e^{-i(-k^z z)} \right) = E^\mu e^{-i(\omega t - k^x x - k^y y)} \left(i k^z e^{(ik^z) z} \right) \\ &= i k^z E^\mu e^{-i(\omega t - \vec{k} \cdot \vec{x})}. \end{aligned}$$

The divergence is then given by:

$$\begin{aligned} \nabla \cdot \vec{\mathcal{E}} &= \frac{\partial \mathcal{E}^x}{\partial x} + \frac{\partial \mathcal{E}^y}{\partial y} + \frac{\partial \mathcal{E}^z}{\partial z} = i k^x E^x e^{-i(\omega t - \vec{k} \cdot \vec{x})} + i k^y E^y e^{-i(\omega t - \vec{k} \cdot \vec{x})} + i k^z E^z e^{-i(\omega t - \vec{k} \cdot \vec{x})} \\ &= i \left(k^x E^x + k^y E^y + k^z E^z \right) e^{-i(\omega t - \vec{k} \cdot \vec{x})} = i \left(\vec{k} \cdot \vec{E} \right) e^{-i(\omega t - \vec{k} \cdot \vec{x})} = i \left(0 \right) e^{-i(\omega t - \vec{k} \cdot \vec{x})} = 0, \end{aligned}$$

and the curl is given by:

$$\begin{aligned} \nabla \times \vec{\mathcal{E}} &= \begin{pmatrix} \frac{\partial \mathcal{E}^z}{\partial y} - \frac{\partial \mathcal{E}^y}{\partial z} \\ \frac{\partial \mathcal{E}^x}{\partial z} - \frac{\partial \mathcal{E}^z}{\partial x} \\ \frac{\partial \mathcal{E}^y}{\partial x} - \frac{\partial \mathcal{E}^x}{\partial y} \end{pmatrix} = i \begin{pmatrix} k^y E^z - k^z E^y \\ k^z E^x - k^x E^z \\ k^x E^y - k^y E^x \end{pmatrix} e^{-i(\omega t - \vec{k} \cdot \vec{x})} \\ &= i(\vec{k} \times \vec{E}) e^{-i(\omega t - \vec{k} \cdot \vec{x})} = (\omega \vec{E}) e^{-i(\omega t - \vec{k} \cdot \vec{x})} = i \left(-i\omega \vec{E} e^{-i(\omega t - \vec{k} \cdot \vec{x})} \right) = i \frac{\partial \vec{\mathcal{E}}}{\partial t}. \end{aligned}$$

Therefore, the plane wave $\vec{\mathcal{E}}$ satisfies the vacuum Maxwell equations. □

0.6 Manifolds

2. *Proof (continuity).* Consider a function $f : \mathbb{R}^n \to \mathbb{R}^m$ We show that f is continuous according to the open pre-image definition if and only if it is according to the epsilon-delta definition.

   First, recall that open n-dimensional and m-dimensional balls are basis elements of $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively.

   Suppose that f satisfies the open pre-image definition: given any open set $U \subseteq \mathbb{R}^m$, the inverse image $f^{-1}U \subseteq \mathbb{R}^n$ is open. Consider some $x \in \mathbb{R}^n$ and $\varepsilon > 0$. Then, there are open sets

   $$V = B_{\mathbb{R}^m}(f(x), \varepsilon) = \{z \in \mathbb{R}^m \mid ||z - f(x)|| < \varepsilon\} \subseteq \mathbb{R}^m \quad \text{and} \quad f^{-1}V = \bigcup_{y \in f^{-1}B} B(y, \delta_y) \subseteq \mathbb{R}^n$$

   with $f(x) \in V$. As a result, there exists $\delta = \delta_x > 0$ such that $x \in B(x, \delta) \subset f^{-1}V$. Thus, $y \in B(x, \delta)$, i.e. $||y - x|| < \delta$, implies that $f(y) \in V$ and thus $||f(y) - f(x)|| < \varepsilon$.

   Now suppose instead that f satisfies the epsilon-delta definition: for all $x \in \mathbb{R}^m$ and all $\varepsilon > 0$, there exists $\delta > 0$ such that $||y - x|| < \delta$ implies $||f(y) - f(x)|| < \varepsilon$. Consider an open set $W \subset \mathbb{R}^m$, and let $W = U \cup V$ for $U \cap f(\mathbb{R}^n) = \emptyset$ and $V \subset f(\mathbb{R}^n)$. Then,

   $$V = \{z \in W \mid z \in f(\mathbb{R}^n)\} = \bigcup_{f(x) \in V} B(f(x), \varepsilon_x) \quad \text{and} \quad f^{-1}V = \bigcup_{f(x) \in V} f^{-1}(B(f(x), \varepsilon_x))$$

   If $x \in f^{-1}V$, then there exists $\delta > 0$ such that $f(x) \in f(B(x, \delta)) \subseteq B(f(x), \varepsilon)) \subseteq V$, so $x \in B(x, \delta)$ and $B(x, \delta) \subseteq f^{-1}V$ yield

   $$f^{-1}V = \bigcup_{x \in f^{-1}V} B(x, \delta_x).$$

   Thus, $f^{-1}W = f^{-1}U \cup f^{-1}V$ is the union of an open set $f^{-1}V$ and an open set $f^{-1}U = \emptyset$.

   □

3. *Proof (sphere manifold).* Let S^n, the n-sphere, be the unit sphere in $\mathbb{R}^{n+1}$

   $$S^n = \{\vec{x} \in \mathbb{R}^{n+1} \mid \sum_{i=1}^{n+1} (x^i)^2 = 1\}$$

   with its induced topology.

   It is easy to prove that S^n is an $(n+1)$-dimensional manifold, so we will show it is in fact an n-dimensional manifold by covering it with two open sets $N, S \subset S^n$

   $$N = U_N = S^n \setminus \{n\} \qquad \text{and} \qquad S = U_S = S^n \setminus \{s\},$$

   where $n = (0, \ldots, +1)$ and $s = (0, \ldots, -1)$ are the north and south poles, respectively. Since $\{n\}, \{s\}$ are singletons, they are closed in $\mathbb{R}^{n+1}$ and in S^n, so their complements N and S in S^n are thus open. We define an atlas $\{\varphi_N : N \to \mathbb{R}^n, \varphi_S : S \to \mathbb{R}^n\}$ via the stereographic projections component-wise for $1 \leq i \leq n$

   $$\varphi_N^i(x) = \frac{x_i}{1 - x_n}, ({\varphi_S}^{-1}(y))^i = \frac{2y_i}{+1 + \sum_{i=1}^n y_i^2} \quad \text{and} \quad \varphi_S^i(x) = \frac{x_i}{1 + x_n}, ({\varphi_S}^{-1}(y))^i = \frac{2y_i}{-1 + \sum_{i=1}^n y_i^2},$$

   and define the $(n+1)^{\text{th}}$ component of the inverse as follows:

   $$({\varphi_N}^{-1}(y))^{n+1} = \frac{-1 + \sum_{i=1}^n y_i^2}{+1 + \sum_{i=1}^n y_i^2}.$$

Notice that the transition function $\varphi_S \circ {\varphi_N}^{-1}$ is smooth and well-defined because it is a composition of smooth operations; the denominator of ${\varphi_n}^{-1}$ is the positive sum of $+1$ and a positive sum of squares, and the denominator of φ_S is only zero on s, on which it is not defined. Similarly, the transition function $\varphi_S \circ {\varphi_N}^{-1}$ is smooth where it is defined.

Therefore, S^n with its induced topology is a manifold. □

4. *Proof (induced manifold).* We show that if M is a manifold with charts $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$ and U is an open subset of M, then U with its induced topology is a manifold.

   We equip U with charts $\psi_\alpha : V_\alpha \to \mathbb{R}^n$, where $V_\alpha = U_\alpha \cap U$ are open sets covering $M \cap U = U$ and where $\psi_\alpha = \varphi_\alpha|_{V_\alpha}$. Then, the transition functions $\psi_\alpha \circ {\psi_\beta}^{-1}$ inherit smoothness via restriction wherever defined. □

5. *Proof (product manifold).* We show that if M and N are m- and n-dimensional manifolds with charts $\varphi_\alpha : U_\alpha \to \mathbb{R}^m$ and $\psi_\beta : V_\beta \to \mathbb{R}^n$ for $\alpha \in A$ and $\beta \in B$, then $M \times N$ is an $(m+n)$-dimensional manifold.

   We can cover the product topology $M \times N$ with open sets $W_\gamma = U_\alpha \times V_\beta$ for $\gamma = \gamma_{\alpha,\beta} \in C = A \times B$, so we define charts $\varphi_\alpha \times \psi_\beta = \xi_\gamma : W_\gamma \to \mathbb{R}^{m+n}$ component-wise. Then, the transition functions $\xi_\gamma \circ {\xi_\delta}^{-1}$ inherit smoothness component-wise wherever defined. □

6. *Proof.* We show that if M and N are n-dimensional manifolds with charts $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$ and $\psi_\beta : V_\beta \to \mathbb{R}^n$ for $\alpha \in A$ and $\beta \in B$, then $M \cup N$ is an n-dimensional manifold.

   We can cover the disjoint union topology $M \cup N$ with open sets $W_\gamma = U_\alpha \cup V_\beta$ for $\gamma = \gamma_{\alpha,\beta} \in C = A \times B$, so we define charts $\xi_\gamma : W_\gamma \to \mathbb{R}^n$ piece-wise by $\xi_\gamma|_{U_\alpha} = \varphi_\alpha$ and $\xi_\gamma|_{V_\beta} = \psi_\beta$. Then, transition functions $\xi_\gamma \circ {\xi_\delta}^{-1}$ inherit smoothness piece-wise wherever defined. □

0.7 Vector Fields

7. *Proof (addition and function multiplication).* We will show that $v+w$ and gw are in $\mathrm{Vect}(M)$ for all $v, w \in \mathrm{Vect}(M)$ and $g \in C^\infty(M)$.

   Let $f, g, h \in C^\infty(M)$, and $\alpha \in \mathbb{R}^n$. Then,

$$
\begin{aligned}
(v+w)(fg) &= v(fg) + w(fg) = v(f)g + fv(g) + w(f)g + fw(g) = (v+w)(f)g + f(v+w)(g),\\
(v+w)(f+g) &= v(f+g) + w(f+g) = v(f) + v(g) + w(f) + w(g) = (v+w)(f) + (v+w)(g)\\
(v+w)(\alpha f) &= v(\alpha f) + w(\alpha f) = \alpha v(f) + \alpha w(f) = \alpha\left(v(f) + w(f)\right) = \alpha(v+w)(f), \text{ and}\\
(gw)(f+h) &= gw(f+h) = gw(f) + gw(h) = (gw)(f) + (gw)(h)\\
(gw)(fh) &= g\left(w(f)h + fw(h)\right) = (gw)(f)h + f(gw)(h)\\
(gw)(\alpha f) &= gw(\alpha f) = \alpha g w(f) = \alpha(gw)(f).
\end{aligned}
$$

   Therefore, $v+w$ and gw are indeed vector fields on M. □

8. *Proof (module of vector fields).* Let $v, w \in \mathrm{Vect}(M)$ and $f, g, h \in C^\infty(M)$. Then, we obtain

$$
\begin{aligned}
(f(v+w))(h) &= f(v+w)(h) = f(v(h) + w(h)) = fv(h) + fw(h)\\
((f+g)v)(h) &= (f+g)v(h) = fv(h) + gv(h)\\
((fg)v)(h) &= (fg)v(h) = f(gv)(h), \text{ and}\\
(1v)(h) &= 1v(h) = v(h).
\end{aligned}
$$

   Therefore, $\mathrm{Vect}(M)$ is a module over $C^\infty(M)$. □

9. *Proof (linear independence).* We show that if $v^\mu \partial_\mu = 0$, we must have $v^\mu = 0$ for all μ.

   Suppose $v^\mu \partial_\mu f = 0$ for all $f \in C^\infty(\mathbb{R}^n)$. Then, $v^\mu = 0$ or $\frac{\partial f}{\partial x^\mu} = 0$ for each $\mu = 1, \ldots, n$. Since the result must hold for all f, which in particular may not satisfy the latter condition, it follows that we must have $v^\mu = 0$ for all μ.

   Therefore, the vector fields $\{\partial_\mu\}$ on $\mathbb{R}^n$ are linearly independent. □

10. *Proof (tangent vector equivalence of vector fields).* We will show that $v = w$ if and only if $v_p = w_p$ for all $p \in M$. If $v = w$, then we obtain

$$v_p(f) = v(f)(p) = w(f)(p) = w_p(f) \qquad \text{for all } f \in C^\infty(M), \tag{9}$$

   for all $p \in M$. If $v_p = w_p$ for all $p \in M$, then we obtain $v = w$ from the following implication:

$$v(f)(p) = v_p(f) = w_p(f) = w(f)(p) \implies v(f) = w(f) \qquad \text{for all } f \in C^\infty(M). \tag{10}$$

   Therefore, every tangent vector at p is of the form v_p for some vector field v. □

11. *Proof (tangent vector space).* We show that T_pM is a vector space over the real numbers.

   Consider scalars $a, b \in \mathbb{R}$ and tangent vectors $u_p, v_p, w_p \in T_pM$ at $p \in M$.

   Then, we show they satisfy the vector space axioms:

$$\begin{aligned}
\text{commutativity} \quad & u_p(f) + v_p(f) = u(f)(p) + v(f)(p) = v(f)(p) + u(f)(p) = v_p(f) + u_p(f) \\
\text{additive identity} \quad & v_p(f) + 0 = v(f)(p) + 0 = v(f)(p) = v_p(f) \\
\text{inverse} \quad & v_p(f) + (-v_p(f)) = v(f)(p) - v(f)(p) = 0 \\
\text{compatibility} \quad & a(bv_p(f)) = a(bv(f)(p)) = (ab)v(f)(p) = (ab)v_p(f) \\
\text{scalar identity} \quad & 1v_p(f) = 1v(f)(p) = v(f)(p) = v_p(f) \\
\text{vector distributivity} \quad & a(u_p(f) + v_p(f)) = a(u(f)(p) + v(f)(p)) = au(f)(p) + av(f)(p) = au_p(f) + av_p(f) \\
\text{field distributivity} \quad & (a+b)v_p(f) = (a+b)v(f)(p) = av(f)(p) + bv(f)(p) = av_p(f) + bv_p(f),
\end{aligned}$$

   and associativity: $u_p(f) + (v_p(f) + w_p(f)) = u(f)(p) + (v(f)(p) + w(f)(p)) = (u(f)(p) + v(f)(p)) + w(f)(p) = (u_p(f) + v_p(f)) + w_p(f)$, for all $f \in C^\infty(M)$. □

12. *Proof (tangent curve).* Consider $\gamma : \mathbb{R} \to M$ and define $\gamma'(t) \in \mathrm{Vect}(M)$ as follows:

$$\gamma' : f \mapsto \frac{d}{dt} f(\gamma(t)).$$

   For some $t \in \mathbb{R}$, let $p = \gamma(t)$ and $\gamma_p = \gamma'(t)$. Then, linearity and the product rule of differentiation yield:

$$\begin{aligned}
\gamma_p(f+g) &= \frac{d}{dt}(f+g)(\gamma(t)) = \frac{d}{dt} f(\gamma(t)) + \frac{d}{dt} g(\gamma(t)) = \gamma_p(f) + \gamma_p(g) \\
\gamma_p(\alpha f) &= \frac{d}{dt}(\alpha f)(\gamma(t)) = \alpha \frac{d}{dt} f(\gamma(t)) = \alpha \gamma_p(f) \\
\gamma_p(fg) &= \frac{d}{dt}(fg)(\gamma(t)) = \frac{d}{dt}\left(f(\gamma(t))\right) g(\gamma(t)) + f(\gamma(t)) \frac{d}{dt} g(\gamma(t)) = \gamma_p(f) g(p) + f(p) \gamma_p(g).
\end{aligned}$$

   Therefore, $\gamma'(t)$ defines a tangent vector at $\gamma(t)$ for each t. □

13. Let $\phi : \mathbb{R} \to \mathbb{R}$ be given by $\phi(t) = e^t$. Let x be the usual coordinate function on $\mathbb{R}$.

Proof. Then, the pullback is given below:

$$\phi^\star x = x(\phi(t)) = x(e^t) = e^x. \tag{11}$$

□

14. Let $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ be rotations counterclockwise by an angle ϕ. Let x, y be the usual coordinate functions on $\mathbb{R}^2$.

    Proof. Then, the pullback is given below:

$$\phi^\star x = x(\phi(x,y)) = x\begin{pmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{pmatrix} = (\cos\theta)x - (\sin\theta)y$$
$$\phi^\star y = y(\phi(x,y)) = y\begin{pmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{pmatrix} = (\sin\theta)x + (\cos\theta)y.$$

□

15. *Proof (smooth map).* Let $f \in C^\infty(N)$, $\gamma : \mathbb{R} \to M$, and $\phi : M \to N$ is smooth.

    If ϕ is smooth, then $\phi \circ \gamma : \mathbb{R} \to N$ is smooth iff $f(\phi \circ \gamma) = \phi^\star f(\gamma)$ depends smoothly on t iff γ is smooth. If ϕ is not smooth, then neither implication holds in general. □

16. *Proof (pushforward of tangent curve).* For any $f \in \mathbb{C}^\infty(M)$, we obtain the following:

$$((\phi \circ \gamma)'(t))(f) = \frac{d}{dt} f(\phi \circ g) = \frac{d}{dt} f \circ \phi \circ g = \frac{d}{dt}\phi^\star f(g) = g'(\phi^\star f) = (\phi_\star g')(f),$$

    where $g = \gamma(t)$ and $g' = \gamma'(t)$ for any $t \in \mathbb{R}$. Therefore, $(\phi \circ \gamma)'(t) = \phi_\star(\gamma'(t))$. □

17. *Proof (pushforward linearity).* Consider the pushforward operation $\phi_\star : T_pM \to T_{\phi(p)}N$ for some $p \in M$ and $\phi : M \to N$. For any $v, w \in T_pM$, $\alpha \in R$ and $f \in C^\infty(N)$, the linearity of T_pM yields the following:

$$\begin{aligned} (\phi_\star(v+w))(f) = (v+w)(\phi^\star f) = v(\phi^\star f) + w(\phi^\star f) &= (\phi_\star v)(f) + (\phi_\star w)(f), \\ (\phi_\star(\alpha v))(f) = (\alpha v)(\phi^\star f) = \alpha v(\phi^\star f) &= \alpha(\phi_\star v)f, \end{aligned}$$

    i.e. that $\phi_\star$ is linear. □

18. Let $\phi : M \to N$ and $v \in \mathrm{Vect}(M)$.

    Proof (pushforward vector field). We compute the pushforward at $f \in C^\infty(N)$ below:

$$\begin{aligned} (\phi_\star v)_q(f) &= (\phi_\star v)(f)(q) && \text{by definition of a tangent vector at } q \\ &= v(\phi^\star f)(p) && \text{by definition of pushforward whenever } \phi(p) = q \\ &= v_p(\phi^\star f) = (\phi_\star v_p)(f) && \text{as above.} \end{aligned}$$

    Therefore, $(\phi_\star v)_q = \phi_\star(v_p)$. □

19. Let $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ be rotation counterclockwise by an angle θ. Let ∂_x, ∂_y be the coordinate vector fields on $\mathbb{R}^2$.

Proof. Let $(x, y) = \phi(\chi, \upsilon) = (\cos(\theta)\chi - \sin(\theta)\upsilon,\ \sin(\theta)\chi + \cos(\theta)\upsilon)$. We evaluate the coordinate vector field pushforwards at $f : \mathbb{R}^2 \to \mathbb{R}$ via the chain rule below.

$$(\phi_\star \partial_x)(f) = \partial_x(\phi^\star f) = \partial_x \left(f \circ \phi \begin{pmatrix} \chi \\ \upsilon \end{pmatrix} \right) = \frac{\partial f}{\partial x}\frac{\partial x}{\partial \chi} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \chi} = \qquad \cos(\theta)\frac{\partial f}{\partial x} + \sin(\theta)\frac{\partial f}{\partial y}$$

$$(\phi_\star \partial_y)(f) = \partial_y(\phi^\star f) = \partial_y f \begin{pmatrix} x \\ y \end{pmatrix} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial \upsilon} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \upsilon} = \qquad -\sin(\theta)\frac{\partial f}{\partial x} + \cos(\theta)\frac{\partial f}{\partial y}$$

□

20. Let v be the vector field $x^2\partial_x + y\partial_y$ on $\mathbb{R}^2$.

Proof. Let $\gamma(t) = (x(t), y(t))$. If $\gamma'(t)(f) = \frac{d}{dt} f(\gamma(t)) = v(f)(\gamma(t)) = v_{\gamma(t)}(f)$ for all $f \in C^\infty(M)$, then we obtain:

$$\gamma'(t)(f) = \left(\frac{\partial f}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial t} \right)(\gamma(t)) = \left(x^2 \frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} \right)(\gamma(t)) = v_{\gamma(t)}(f).$$

From $\dot{x}(t) = x(t)^2$ and $\dot{y}(t) = y(t)$, we obtain $x(t) = \frac{1}{c_1 - t}$ and $y(t) = c_2 e^t$ as general solutions with $c_1 = \frac{1}{x(0)} + 0$ and $c_2 = \frac{y(0)}{e^0} = y(0)$, which yields the following particular solution:

$$\gamma(t) = \begin{pmatrix} (1/x(0) - t)^{-1} \\ y(0)e^t \end{pmatrix},$$

which is defined whenever $\frac{1}{x(0)} - t \neq 0$, which holds for all t iff $x(0) = 0$.

Thus, the integral curves $\gamma(t)$ defined for all t are those which cross the origin. □

21. Let $s, r \in \mathbb{R}$.

Proof (flow semigroup). Let $p \in M$ and $\gamma_p : \mathbb{R} \to M$ be the integral curve through $\gamma_p(0) = p$ defined by $\gamma_p : s \mapsto \phi_s(p)$. Then, ϕ_0 is the identity because $\phi_0(p) = \gamma_p(0) = p$. Furthermore, we obtain $\phi_t \circ \phi_s = \phi_{t+s}$ as follows: $\phi_{t+s}(p) = \gamma_p(t+s) = \gamma_{\gamma_s(p)}(t) = \phi_t(\phi_s(p))$. □

22. Consider the normalized vector fields in the r and θ directions on the plane in polar coordinates (not defined at the origin):

$$v = \frac{x\partial_x + y\partial_y}{\sqrt{x^2 + y^2}} \tag{12}$$

$$w = \frac{x\partial_x - y\partial_y}{\sqrt{x^2 + y^2}} \tag{13}$$

Proof. Let $r = \sqrt{x^2 + y^2}$. Then, the commutativity of the ∂_μ yield the following commutator:

$$\begin{aligned}
[v, w] &= \left(\frac{x\partial_x + y\partial_y}{r} \right)\left(\frac{x\partial_x - y\partial_y}{r} \right) - \left(\frac{x\partial_x - y\partial_y}{r} \right)\left(\frac{x\partial_x + y\partial_y}{r} \right) \\
&= \frac{1}{r^2}\left(x\partial_x(x\partial_x - y\partial_y) + y\partial_y(x\partial_x - y\partial_y) - x\partial_x(x\partial_x + y\partial_y) + y\partial_y(x\partial_x + y\partial_y) \right) \\
&= \frac{0 + 0 + 0 + 0}{r^2} = 0.
\end{aligned}$$

□

23. If $v, w \in \mathrm{Vect}M$ generate the flows $\phi_t, \psi_s : M \to M$, then $f \in C^\infty(M)$ gives the following.

$$(vf)(p) = \frac{d}{dt} f(\phi_t(p))\Big|_{t=0} \qquad \text{and} \qquad (wf)(p) = \frac{d}{ds} f(\psi_s(p))\Big|_{s=0}$$

Proof (commutator flow). Since $\phi(p) : \mathbb{R} \to M$ defines a curve $t \mapsto \phi_t(p) = \gamma_p(t)$, we use the tangent vector $\phi_t'(p)(f) = \left(\frac{d}{dt}\phi_t(p)\right)(f) = v_{\phi_t(p)}(f) = v_{\gamma(t)}(f) = \gamma'(t)(f)$ to compute the composite flows below:

$$\begin{aligned} v(w(f)) &= \frac{d}{dt} wf(\phi_t)\Big|_{t=0} = \frac{d}{dt}\phi_t(wf)\Big|_{t=0} = \frac{d}{dt}\phi_t\left(\frac{d}{ds} f(\psi_s)\Big|_{s=0}\right)\Big|_{t=0} \\ &= \frac{d}{dt}\phi_t\left(\frac{d}{ds}\psi_s(f)\Big|_{s=0}\right)\Big|_{t=0} = \frac{d}{dt}\frac{d}{ds}\psi_s(f(\phi_t))\Big|_{s=0,t=0} = \frac{d^2}{dtds} f(\phi_t(\psi_s))\Big|_{s=t=0}, \end{aligned}$$

and $w(v(f)) = \dfrac{d^2}{dtds} f(\psi_s(\phi_t))\Big|_{s=t=0}$ follows from symmetry.

We apply subtraction and evaluation to check the following equation:

$$[v, w](f)(p) = (vwf)(p) - (wvf)(p) = \frac{d^2}{dtds} f(\phi_t(\psi_s(p))) - f(\psi_s(\phi_t(p)))\Big|_{s=t=0}. \tag{14}$$

□

24. Let $u, v, w \in \mathrm{Vect}(M)$ and $\alpha, \beta \in \mathbb{R}$.

Proof (commutator identities). Let v, w be as above and let $(uf)(p) = \dfrac{d}{dt} f(\chi_r(p))\,|_{r=0}$. Then, we obtain anti-commutativity and linearity as follows:

$$\begin{aligned} -[w, v](f) &= -\frac{d^2}{dtds}\left[f(\psi_s(\phi_t)) - f(\phi_t(\psi_s))\right]_{s=t=0} = [v, w] \\ [u, \alpha v](f) &= \alpha\frac{d^2}{drdt} f(\chi_r(\phi_t(p))) - \alpha\frac{d^2}{drdt} f(\phi_t(\chi_r(p))) = \alpha[u, v](f) \\ [u, v + w](f) &= \frac{d^2}{drdt} f(\chi_r((\phi_t + \psi_t)(p))) - \frac{d^2}{drdt} f((\phi_t + \psi_t)(\chi_r(p))) \\ &= \frac{d^2}{drdt}\left(f(\chi_r(\phi_t(p))) + f(\chi_r(\psi_t(p))) - f(\phi_t(\chi_r(p))) - f(\psi_t(\chi_r(p)))\right) = [u, v](f) + [u, w](f) \\ [u, \alpha v + \beta w] &= \alpha[u, v] + \beta[u, w]. \end{aligned}$$

We compute $[u, [v, w]](f) + [v, [w, u]](f) + [w, [u, v]](f) = \dfrac{d^3}{drdtds} 0\Big|_{s=t=r=0} = 0$ below:

$$\begin{aligned} \frac{d^3}{drdtds}\Big[& f(\chi_r(\phi_t(\psi_s))) - f(\phi_t(\psi_s(\chi_r))) - f(\chi_r(\psi_s(\phi_t))) + f(\psi_s(\phi_t(\chi_r))) \\ & f(\phi_t(\psi_s(\chi_r))) - f(\psi_s(\chi_r(\phi_t))) - f(\phi_t(\chi_r(\psi_s))) + f(\chi_r(\psi_s(\phi_t))) \\ & f(\psi_s(\chi_r(\phi_t))) - f(\chi_r(\phi_t(\psi_s))) - f(\psi_s(\phi_t(\chi_r))) + f(\phi_t(\chi_r(\psi_s)))\Big]_{s=t=r=0}. \end{aligned}$$

□

0.8 Differential Forms

25. *Proof (1-form linearity).* Let $\omega, \mu \in \Omega^1(M)$, $f, g \in C^\infty(M)$, and $v, w \in \text{Vect}(M)$.
We show linearity over $C^\infty(M)$ below:

$$\begin{aligned}(\omega+\mu)(v+w) &= \omega(v+w)+\mu(v+w) = \omega(v)+\omega(w)+\mu(v)+\mu(w) = (\omega+\mu)(v)+(\omega+\mu)(w)\\ (\omega+\mu)(gv) &= \omega(gv)+\mu(gv) = g\omega(v)+g\mu(v) = g(\omega+\mu)(v)\\ (f\omega)(v+w) &= f\omega(v+w) = f\omega(v)+f\omega(w) = (f\omega)(v)+(f\omega)(w)\\ (f\omega)(gv) &= f(\omega(gv)) = (fg)\omega(v) = (gf)\omega(v) = g(f\omega(v)).\end{aligned}$$

Therefore, $\omega+\mu$ and $f\omega$ really are 1-forms. □

26. *Proof (1-form module).* Let $\omega, \mu \in \Omega^1(M)$ and $f, g, \omega(v), \mu(v) \in C^\infty(M)$ for some $v \in \text{Vect}(M)$.
We apply 1-form linearity and commutative algebra rules below:

$$\begin{aligned}f(\omega+\mu)(v) &= f[\omega(v)+\mu(v)] = f\omega(v)+f\mu(v)\\ (f+g)\mu(v) &= f\mu(v)+g\mu(v)\\ (fg)\mu(v) &= f(g\mu(v))\\ 1\mu(v) &= \mu(v).\end{aligned}$$

This shows that $\Omega^1(M)$ is a module over $C^\infty(M)$. □

27. Let $f, g, h \in C^\infty(M)$ and $\alpha \in \mathbb{R}$.

Proof (linear Leibniz differential). The properties of a vector field $v \in \text{Vect}(M)$ yield:

$$\begin{aligned}d(f+g)(v) &= v(f+g) = v(f)+v(g) = df(v)+dg(v)\\ d(\alpha f)(v) &= v(\alpha f)\alpha v(f) = \alpha\ df\\ (f+g)dh(v) &= (f+g)v(h) = fv(h)+gv(h) = f\ dh+g\ dh\\ d(fg)(v) &= v(fg) = v(f)g+fv(g) = f\ dg+g\ df.\end{aligned}$$

Therefore, the exterior derivative obeys Leibniz and linearity laws. □

28. Suppose $f(x^1, \ldots, x^n)$ is a function on $\mathbb{R}^n$.

Proof (equivalence of differential and gradient). For any $v \in \text{Vect}(M)$, the span of $\mathbb{R}^n$ by $\{\partial_\mu\}$ yields:

$$df(v) = vf = \nabla f \cdot (v^\nu \partial_\nu) = \partial_\mu f v^\nu \delta^\mu_\nu = \partial_\mu f dx^\mu(v^\nu\partial_\nu) = \partial_\mu f dx^\mu(v).$$

□

29. Suppose $\omega = \omega_\mu dx^\mu = 0$ for some $\omega \in \Omega^1(M)$.

Proof (1-forms linearly independent). Since $\omega(v) = \omega_\nu v^\nu = 0$, it must be that $\omega_\nu = 0$ or $v^\nu = 0$ for each ν. Since this must hold for all v, it follows that each function $\omega_\nu = 0$. □

30. *Proof (cotagent vectors well-defined).* Let $\omega \in \Omega^1(M)$ and $v, w \in \text{Vect}(M)$. If $v_p = w_p$, then $v(v)(p) = \omega(v)(p)$ even though it may be the case that $v_q \neq w_q$ for some $p, q \in M$.
If $\omega_p = \nu_p$ for all $p \in M$, then $\nu_p(v_p) = \omega_p(v_p) = \omega(v)(p) = \nu(v)(p)$ implies $\omega(v) = \nu(v)$ for all $v \in \text{Vect}(M)$. Therefore, $\omega = \nu$. □

31. Suppose that we have vector spaces with linear maps $f : V \to W$ and $g : W \to X$.

Proof (cotangent contravariance). First, we show that the dual $1_V^\star$ of the identity map 1_V on V is the identity map $1_{V^\star}$ on $V^\star$:

$$(1_V^\star \omega)(v) = \omega(1_V(v)) = \omega(v) = 1_{V^\star}(\omega(v)).$$

For any $\omega \in V^\star$, we also obtain:

$$(gf)^\star(\omega) = \omega(g(f)) = \omega \circ g(f) = f^\star(\omega \circ g) = f^\star(\omega(g)) = f^\star(g^\star \omega) = f^\star g^\star(\omega),$$

which shows that cotangent vectors are contravariant. □

32. *Proof (1-form pullback).* Let on $v \in \mathrm{Vect}(N)$ and $f \in C^\infty(N)$
The pullback of $\omega \in \Omega^1(N)$ by $\phi : M \to N$ is given by $\phi^\star \omega \in \Omega^1(M)$ below:

$$(\phi^\star \omega)(v)(f) = \omega(\phi_\star v(f)) = \omega(v(\phi^\star f)) = \omega(v(f \circ \phi)),$$

which is well-defined since $f \circ \phi \in C^\infty(M)$ and $v \circ \phi^\star \in \mathrm{Vect}(M)$, $\omega \circ v \circ \phi^\star \in C^\infty(M)$.
Notice that $\phi^\star \omega = \psi^\star \omega$ implies $\psi = \phi$. Thus, the pullback of 1-forms exists and is unique. □

33. Let $\phi : \mathbb{R} \to \mathbb{R}$ be given by $\phi(t) = \sin(t)$. Let dx be the usual 1-form on $\mathbb{R}$.

Proof. We show that $\phi_\star dx = \cos t \; dt$ as follows:

$$\phi_\star dx = dx(\phi^\star) = \phi^\star(dx) = dx(\phi) = d(\sin t) = \cos t \; dt.$$

□

34. Let $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ denote rotation counterclockwise by the angle θ. Let dx, dy be the usual basis of 1-forms on $\mathbb{R}^2$.

Proof. We compute the pullback of the basis 1-forms below:

$$\phi^\star dx = dx(\phi(x, y)) = dx \begin{pmatrix} \cos\theta x - \sin\theta y \\ \sin\theta x + \cos\theta y \end{pmatrix} = d(\cos\theta x - \sin\theta y) = \cos\theta \cdot 1 \; dx - \sin\theta \cdot 1 \; dy$$

$$\phi^\star dy = d(\sin\theta x + \cos\theta y) = \sin\theta \; dx + \cos\theta \; dy.$$

□

35. *Proof (coordinate 1-forms).* We apply naturality of 1-forms $dx^\mu = \varphi^\star dx^\mu$ on U below:

$$\varphi^\star dx^\mu = d(\varphi^\star x^\mu),$$

and thus obtain the differentials $d(\varphi^\star x^\mu)$ of the local coordinates on U. □

36. *Proof (change of coordinate 1-forms).* We apply a change of coordinates to $v'^\nu = dx'^\nu$ below:

$$dx'^\nu = v'^\nu = \frac{\partial x'^\nu}{\partial x^\mu} v^\mu = \frac{\partial x'^\nu}{\partial x^\mu} dx^\mu.$$

If we can express any 1-form ω on $\mathbb{R}^n$ in your components ω'_ν and my components ω_μ as follows:

$$\omega = \omega_\mu dx^\mu = \omega'_\nu dx'^\nu,$$

then applying the definitions and equating coefficients yields the following relation:

$$\omega'_\nu = \omega_\mu \frac{dx^\mu}{dx'^\nu}.$$

□

37. *Proof (pullback coordinates).* We pull the coordinate 1-form dx'^ν back by ϕ below:

$$(\phi^\star dx'^\nu)(\partial_\lambda) = dx'^\nu(\phi_\star \partial_\lambda) = dx'^\nu\left(\frac{\partial x'^\nu}{\partial x^\lambda}\partial'_\nu\right) = \frac{\partial x'^\nu}{\partial x^\lambda}\left(dx'^\nu \partial'_\nu\right) = \frac{\partial x'^\nu}{\partial x^\mu}\delta^\mu_\lambda = \frac{\partial x'^\nu}{\partial x^\mu} dx^\mu(\partial_\lambda),$$

and obtain $\dfrac{\partial x'^\nu}{\partial x^\mu}$ from the left and the right. □

38. Let

$$e_\mu = T^\nu_\mu \partial_\nu,$$

where ∂_ν are the coordinate vector fields associated to local coordinates on an open set U, and T^ν_μ are functions on U.

Proof (transformed basis). We will show that the vector fields e_μ are a basis of vector fields on U if ($\Leftarrow$) and only if ($\Rightarrow$) for each $p \in U$ the matrix $T^\nu_\mu(p)$ is invertible.

Let $v \in \mathrm{Vect}(U)$, and recall that $v = v^\mu \partial_\mu$ for some $v^\mu \in C^\infty(U)$.

$\Rightarrow$: Suppose the matrices $T^\nu_\mu(p)$ are invertible for each $p \in U$. Then, we obtain:

$$v = (T^\nu_\mu)^{-1} T^\nu_\mu v^\mu \partial_\mu = (T^\nu_\mu)^{-1} v^\mu T^\nu_\mu \partial_\mu = (T^\nu_\mu)^{-1} v^\mu e_\mu,$$

which shows that the e_μ are a basis of vector fields on U.

$\Leftarrow$: Suppose the vector fields e_μ are a basis of vector fields on U. Then, $v = u^\mu e_\mu = u^\mu T^\nu_\mu \partial_\nu = v^\mu \partial_\nu$ for some $u^\mu \in C^\infty(U)$, so $u^\mu T^\nu_\mu = v^\mu$. Since this holds for $v^\mu = 1$, there must exist $u^\mu = (T^\nu_\mu)^{-1}$. Thus, the matrices $T^\nu_\mu(p)$ are invertible for each $p \in U$.

□

39. *Proof (dual basis).* We seek $f^\mu \in \Omega^1(M)$ satisfying $f^\mu(e_\nu) = f^\mu(T^\mu_\nu \partial_\mu) = \delta^\mu_\nu$.

Consider $f^\mu = (T^\mu_\nu)^{-1} dx^\nu$. Notice that $f^\mu(e_\nu) = (T^\mu_\nu)^{-1} dx^\nu(T^\mu_\nu \partial_\mu) = 1 \cdot dx^\nu(\partial_\mu) = \delta^\mu_\nu$.

Suppose $g^\mu \in \Omega^1(M)$ satisfies $g^\mu(e_\nu) = \delta^\mu_\nu$. Then, $g^\mu = f^\mu$. □

40. Let e_μ be a basis of vector fields on U and let f^μ be the dual basis of 1-forms. Let

$$e'_\mu = T^\nu_\mu e_\nu$$

be another basis of vector fields, and let f'^μ be the corresponding dual basis of 1-forms.

Proof (change of coordinates). Since $f'^\mu(e'_\mu) = \delta^\mu_\nu = (T^{-1})^\mu_\nu f^\nu(T^\nu_\mu e_v)$, it follows that $f'^\mu = (T^{-1})^\mu_\nu f^\nu$. If $v = v^\mu e_\mu = v'^\mu e'_\mu$, then evaluation with the identity above yields:

$$v'^\mu = (T^{-1})^\mu_\nu v^\nu,$$

If $\omega = \omega_\mu f^\mu = w'_\mu f'^\mu$, then evaluation with the identity above yields:

$$\omega'_\mu = T^\nu_\mu \omega_\nu.$$

□

41. *Proof.* Let $u = u_x dx + u_y dy + u_z dz$, $v = v_x dx + v_y dy + v_z dz$, and $w = w_x dx + w_y dy + w_z dz$. We compute the 3-form $u \wedge v \wedge w$ below:

$$\begin{aligned}
&(u_x dx + u_y dy + u_z dz) \wedge (v_x dx + v_y dy + v_z dz) \wedge (w_x dx + w_y dy + w_z dz) \\
&= u_x dx \wedge (v_y dy + v_z dz) \wedge (w_y dy + w_z dz) \\
&\quad + u_y dy \wedge (v_x dx + v_z dz) \wedge (w_x dx + w_z dz) \\
&\quad + u_z dz \wedge (v_x dx + v_y dy) \wedge (w_x dx + w_y dy) \\
&= \left[u_x(v_y w_z - v_z w_y) + u_y(v_x w_z - w_z v_x) + u_z(v_x w_y - w_x v_y)\right] dx \wedge dy \wedge dz \\
&= \det \begin{pmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{pmatrix} dx \wedge dy \wedge dz.
\end{aligned}$$

It is an example of the following operation:

$$\begin{aligned}
\vec{u} \cdot (\vec{v} \times \vec{w}) &= \begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix} \cdot \begin{pmatrix} v_y w_z - v_z w_y \\ v_z w_x - v_x w_z \\ v_x w_y - v_y w_x \end{pmatrix} \\
&= u_x(v_y w_z - v_z w_y) + u_y(v_z w_x - v_x w_z) + u_z(v_x w_y - v_y w_x).
\end{aligned}$$

□

42. *Proof.* Let a, b, c, d be four vectors in a 3-dimensional space V with a basis $\{dx^\mu\}_{\mu=1}^3$ for some $d = d_1 dx^1 + d_2 dx^2 + d_3 dx^3$, and let $a \wedge b \wedge c = e dx^1 \wedge dx^2 \wedge dx^3$. Then, we compute $a \wedge b \wedge c \wedge d$:

$$\begin{aligned}
e \wedge d &= e\left[d_1(dx^1 \wedge dx^1) \wedge dx^2 \wedge dx^3 + d_2 dx^1 \wedge (dx^2 \wedge dx_2) \wedge dx^3 + d_3 dx^1 \wedge dx^2 \wedge (dx^3 \wedge dx^3)\right] \\
&= e\left[0 d_1 dx^2 \wedge dx^3 + 0 d_2 dx^1 \wedge dx^3 + 0 d_3 dx^1 \wedge dx^2\right] = e[0 + 0 + 0] = 0e = 0.
\end{aligned}$$

□

43. *Proof.* If V is 1-dimensional and thus spanned by some dx, then ΛV consists of 1 and dx; note that $dx \wedge dx = 0$ and similarly for $1 < p$-form wedges.

If V is 2-dimensional, then ΛV consists of $1, dx_1, dx_2$ and $dx_1 \wedge dx_2 = -dx_2 \wedge dx_1$; note that $dx_1 \wedge dx_2 \wedge dx_\mu = 0$ and similarly for $2 < p$-form wedges.

If V is 4-dimensional, then ΛV consists of $1, dx_1, dx_2, dx_3, dx_4$, and all possible wedges of unique pairs or triplets of dx_μ; note that any wedge product with duplicate 1-forms, and thus any 5-fold wedge, is equal to 0. □

44. *Sketch (dimension of p-fold products of vectors).* Let V be an n-dimensional vector space. Notice that $\Lambda^p V$ is empty for $p > n$ follows from the pigeonhole principle and the fact that $v \wedge v = 0$. For $0 \leq p \leq n$, the dimension of Λ^V corresponds to the number of choices in $v = \bigwedge_{i=1}^p v_i$ which is $\frac{n!}{p!(n-p)!}$ up to permutations, since reorderings are equivalent by anti-commutativity. □

45. *Sketch (direct sum of subspaces).* We aim to show that ΛV is the direct sum of the subspaces $\Lambda^p V$:

$$\Lambda V = \bigoplus \Lambda^p V,$$

and that the dimension of ΛV is 2^n if V is n-dimensional.

Indeed, any basis element of ΛV is a p-form for some p, which can be expressed as an n-tuple whose p^{th} and no other is non-zero. Further, elements of ΛV can be thought of as subsets of the basis elements v_i: either on or off; there are 2^n such subsets. □

46. *Sketch (graded commutative forms).* Given a vector space V, we show that ΛV is a graded commutative or supercommutative algebra, that is, if $\omega \in \Lambda^p V$ and $\mu \in \Lambda^q V$, then

$$\omega \wedge \mu = (-1)^{pq} \mu \wedge \omega,$$

and that $\Omega(M)$ is graded commutative for any manifold M.

This follows from realizing ω and μ as a wedge of p- and q-fold products of vectors and anti-commutative the left-most vector of μ until it is further left than the left-most vector of ω. Each transposition adds a factor of -1, and there are pq transpositions in total. The same argument holds for differential forms, with 1-forms taking the place of vectors. □

47. Show that differential forms are contravariant. That is, show that if $\phi : M \to N$ is a map from the manifold M to the manifold N, there is a unique pullback map

$$\phi^\star : \Omega(N) \to \Omega(M)$$

agreeing with the usual pullback on 0-forms (functions) and 1-forms, and satisfying

$$\begin{aligned} \phi^\star(\alpha\omega) &= \alpha\phi^\star(\omega) \\ \phi^\star(\omega + \mu) &= \phi^\star\omega + \phi^\star\mu \\ \phi^\star(\omega \wedge \mu) &= \phi^\star\omega \wedge \phi^\star\mu. \end{aligned}$$

Guess (N/A). Seems important, but will skip for now. Intend to actually think about and maybe discuss on Friday. Maybe just $\phi^\star\omega = \omega(\phi^\star)$ if that type checks. □

48. Compare how 1-forms and 2-forms on $\mathbb{R}^3$ transform under parity. That is, let $P : \mathbb{R}^3 \to \mathbb{R}^3$ be the map

$$P(x, y, z) = (-x, -y, -z),$$

known as the 'parity transformation'. Note that P maps right-handed bases to left-handed bases and vice versa. Compute $\phi^\star(\omega)$ when ω is the 1-form $\omega_\mu dx^\mu$, and when it is the 2-form $\frac{1}{2}\omega_{\mu\nu} dx^\mu \wedge dx^\nu$.

Guess (N/A). Need to do previous problem first :(□

49. Show that on $\mathbb{R}^n$ the exterior derivative of any 1-form is given by

$$d(\omega_\mu dx^\mu) = \partial_\nu \omega_\mu dx^\nu \wedge dx^\mu$$

.

Guess (N/A). Intend to do. □